

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12419042
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Hong; Jiseok et al.

Integrated circuit device

Abstract

An integrated circuit device includes: a substrate including a plurality of active regions; a bit line extending on the substrate in a horizontal direction; a direct contact connected between a first active region selected among the plurality of active regions and the bit line; an inner oxide layer contacting a sidewall of the direct contact; and a carbon-containing oxide layer nonlinearly extending on a sidewall of the bit line in a vertical direction, the carbon-containing oxide layer contacting the sidewall of the bit line.

Inventors: Hong; Jiseok (Suwon-si, KR), Lee; Sangho (Seoul, KR), Park; Seoryong (Ansan-si, KR), Ahn; Jiyoung (Seoul, KR), Lee; Kiseok (Seoul, KR), Lee; Kiseok (Hwaseong-si, KR), Choi; Yoonyoung (Seoul, KR), Han; Seunguk (Suwon-si, KR)

Applicant: SAMSUNG ELECTRONICS CO., LTD. (Suwon-si, KR)

Family ID: 1000008819901

Assignee: SAMSUNG ELECTRONICS CO., LTD. (Suwon-si, KR)

Appl. No.: 18/186593

Filed: March 20, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20230232616 A1	Jul. 20, 2023

Foreign Application Priority Data

KR	10-2022-0076763	Jun. 23, 2020
----	-----------------	---------------

Related U.S. Application Data

Publication Classification

Int. Cl.: H10B12/00 (20230101)

U.S. Cl.:

CPC H10B12/485 (20230201); H10B12/0335 (20230201); H10B12/315 (20230201); H10B12/482 (20230201);

Field of Classification Search

CPC: H01L (27/10814); H01L (27/10855); H01L (27/10885); H01L (27/10888); H10B (12/315); H10B (12/482); H10B (12/485)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
8643087	12/2013	Sandhu et al.	N/A	N/A
8969936	12/2014	Lee et al.	N/A	N/A
9177891	12/2014	Kim et al.	N/A	N/A
9721964	12/2016	Lee	N/A	N/A
10157841	12/2017	Borsari	N/A	H10B 12/315
10461153	12/2018	Lee et al.	N/A	N/A
10522548	12/2018	Chun	N/A	N/A
10573652	12/2019	Lee	N/A	H10B 12/315
10847519	12/2019	Ji	N/A	H01L 21/02126
2015/0187644	12/2014	Kim	N/A	N/A
2017/0278848	12/2016	Wu	N/A	N/A
2018/0175040	12/2017	Kim	N/A	H01L 21/76224
2019/0096890	12/2018	Lee	N/A	H10B 12/485
2019/0296024	12/2018	Ji et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
10-2004-0050799	12/2003	KR	N/A
10-2014-0052225	12/2013	KR	N/A
10-2019-0035250	12/2018	KR	N/A
10-2019-0112444	12/2018	KR	N/A
201546966	12/2014	TW	N/A

OTHER PUBLICATIONS

Communication issued Jun. 19, 2024 by the Korean Patent Office for KR Patent Application No. 10-2020-0076763. cited by applicant

Primary Examiner: Page; Steven M

Attorney, Agent or Firm: Sughrue Mion, PLLC

Background/Summary

CROSS-REFERENCE TO THE RELATED APPLICATION (1) This application is a Continuation of U.S. patent application Ser. No. 17/168,952, filed on Feb. 5, 2021, which is based on and claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2020-0076763, filed on Jun. 23, 2020, in the Korean Intellectual Property Office, the disclosure of which is incorporated by reference herein in its entirety.

BACKGROUND

(1) The inventive concept relates to an integrated circuit device, and more particularly, to an integrated circuit device including a plurality of conductive lines.

(2) Recently, as down-scaling of integrated circuit devices has progressed rapidly, spaces between each of a plurality of conductive lines are reduced, and accordingly, separation distances between the plurality of conductive lines and between each of the plurality of conductive regions have gradually decreased. Accordingly, there is a need to develop a technology for implementing an integrated circuit device capable of suppressing parasitic capacitance between the plurality of conductive lines and other conductive regions adjacent thereto and maintaining a structure in which the plurality of conductive lines are stable and reliable.

SUMMARY

(3) An aspect of the inventive concept is to provide an integrated circuit device capable of suppressing parasitic capacitance between a conductive line and another conductive line adjacent thereto even when an area of a device region is reduced according to down-scaling of a semiconductor device, and maintaining a structure in which the conductive line is stable and reliable.

(4) According to an aspect of the disclosure, there is provided an integrated circuit device comprising: a substrate comprising a plurality of active regions; a bit line extending on the substrate in a first direction; a direct contact connected between a first active region among the plurality of active regions and the bit line; an inner oxide layer contacting a sidewall of the direct contact; and a carbon-containing oxide layer extending on a sidewall of the bit line in a second direction perpendicular to the first direction, the carbon-containing oxide layer contacting the sidewall of the bit line.

(5) According to another aspect of the disclosure, there is provided an integrated circuit device comprising: a substrate comprising a plurality of active regions; a plurality of bit lines spaced apart from each other on the substrate in a first direction, the plurality of bit lines extending in a second direction crossing the first direction; a direct contact connected provided between a first active region among the plurality of active regions and a first bit line among the plurality of bit lines; a contact plug connected to a second active region adjacent to the first active region among the plurality of active regions, the contact plug extending on the substrate in a third direction perpendicular to the first direction and the second direction; and a spacer structure provided between the first bit line and the contact plug, wherein the spacer structure comprises: an inner oxide layer contacting a sidewall of the direct contact; and a carbon-containing oxide layer extending on a sidewall of the first bit line in the third direction, the carbon-containing oxide layer

being in direct contact with a sidewall of the first bit line.

(6) According to another aspect of the disclosure, there is provided an integrated circuit device comprising: a substrate comprising a plurality of active regions spaced apart from each other; a first bit line and a second bit line spaced apart from each other on the substrate in a first direction, the first bit line and the second bit line extending in a second direction crossing the first direction; a plurality of contact plugs arranged in a row between the first bit line and the second bit line in the second direction; a plurality of insulating fences provided respectively between the plurality of contact plugs; a direct contact connected between a first active region among the plurality of active regions and the first bit line; and a spacer structure provided between the first bit line and the contact plug, wherein the spacer structure comprises: an inner oxide layer contacting a sidewall of the direct contact, the inner oxide layer comprising a silicon oxide layer; and an SiOC layer extending on a sidewall of the first bit line in a third direction, the SiOC layer contacting the sidewall of the first bit line.

(7) According to another aspect of the disclosure, there is provided an integrated circuit device comprising: a substrate comprising a plurality of active regions; a bit line extending on the substrate in a first direction, the bit line comprising a lower conductive layer, an intermediate conductive layer, and an upper conductive layer sequentially stacked on the substrate; the lower conductive layer including a doped polysilicon layer; an inner oxide layer contacting a first portion of a sidewall of the bit line at the lower conductive layer; and a carbon-containing oxide layer contacting a second portion of the sidewall above the first portion of the sidewall.

(8) According to another aspect of the disclosure, there is provided an integrated circuit device comprising: a substrate comprising a plurality of active regions; a bit line extending on the substrate in a first direction; a direct contact having a lower surface contacting a first active region among the plurality of active regions and an upper surface contacting the bit line, the direct contact including a doped polysilicon layer; an inner oxide layer contacting a sidewall of the direct contact at a lower portion of the direct contact; and a carbon-containing oxide layer contacting the sidewall of the direct contact at an upper portion of the direct contact and contacting the sidewall of the bit line.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Embodiments of the inventive concept will be more clearly understood from the following detailed description taken in conjunction with the accompanying drawings in which:

(2) FIG. 1 is a schematic planar layout of main components of a memory cell array region of an integrated circuit device, according to an example embodiment of the inventive concept;

(3) FIGS. 2A, 2B and 2C are cross-sectional views of an integrated circuit device according to an example embodiment of the inventive concept;

(4) FIGS. 3 through 10 are cross-sectional views of integrated circuit devices according to other example embodiments of the inventive concept;

(5) FIGS. 11A through 11O are cross-sectional views illustrating a manufacturing method of an integrated circuit device according to process sequences, according to example embodiments of the inventive concept;

(6) FIGS. 12A through 12H are cross-sectional views illustrating a manufacturing method of an integrated circuit device according to process sequences, according to another example embodiments of the inventive concept; and

(7) FIGS. 13A and 13B are cross-sectional views illustrating a manufacturing method of an integrated circuit device according to process sequences, according to another example embodiments of the inventive concept.

DETAILED DESCRIPTION OF THE EMBODIMENTS

(8) Hereinafter, example embodiments of the inventive concept will be described in detail with reference to the accompanying drawings. Identical reference numerals are used for the same constituent devices in the drawings, and a duplicate description thereof will be omitted.

(9) FIG. 1 is a schematic planar layout of a memory cell array area of an integrated circuit device **10**, according to an example embodiment of the inventive concept.

(10) Referring to FIG. 1, the integrated circuit device **10** may include a plurality of active regions ACT. The plurality of active regions ACT may be arranged in a diagonal direction with respect to a first horizontal direction (X direction) and a second horizontal direction (Y direction).

(11) A plurality of word lines WL may extend in parallel with each other in the first horizontal direction (X direction) across the plurality of active regions ACT. On the plurality of word lines WL, a plurality of bit lines BL may extend in parallel with each other in the second horizontal direction (Y direction) across the first horizontal direction (X direction). The plurality of bit lines BL may be connected to the plurality of active regions ACT via direct contacts DC.

(12) A plurality of buried contacts BC may be between two adjacent bit lines BL among the plurality of bit lines BL. According to an example embodiment, the plurality of buried contacts BC may be arranged in a line in the first horizontal direction (X direction) and the second horizontal direction (Y direction), respectively. A plurality of conductive landing pads LP may be formed on the plurality of buried contacts BC. The plurality of buried contacts BC and the plurality of conductive landing pads LP may connect bottom electrodes of capacitors formed on top portions of the plurality of bit lines BL to the active region ACT. At least a portion of each of the plurality of conductive landing pads LP may vertically overlap the buried contact BC.

(13) Next, example configurations of integrated circuit devices according to embodiments of the inventive concept are described with reference to FIGS. 2 through 10. Each of the integrated circuit devices illustrated in FIGS. 2 through 10 may have a layout of the integrated circuit device **10** illustrated in FIG. 1 according to various example embodiments.

(14) FIGS. 2A-2C are cross-sectional views of an integrated circuit device **100** according to an example embodiment of the inventive concept. FIG. 2A is a cross-sectional view of some components of a portion corresponding to a cross-section taken along line A-A' of FIG. 1, FIG. 2B is a cross-sectional view of some components of a portion corresponding to a cross-section taken along line B-B' of FIG. 1, and FIG. 2C is an enlarged cross-sectional view of a portion corresponding to a dash-lined region AX in FIG. 2A.

(15) Referring to FIGS. 2A-2C, the integrated circuit device **100** may include a substrate **110** in which a plurality of active regions ACT are defined by a device isolation layer **112**. The device isolation layer **112** may be in a device isolation trench T1 in the substrate **110**.

(16) According to an example embodiment, the substrate **110** may include silicon, for example, monocrystalline silicon, polycrystalline silicon, or amorphous silicon. According to another example embodiment, the substrate **110** may include at least one of Ge, SiGe, SiC, GaAs, InAs, or InP. According to an example embodiment, the substrate **110** may include conductive regions, for example, a well doped with an impurity, or a structure doped with an impurity. The device isolation layer **112** may include an oxide layer, a nitride layer, or a combination thereof.

(17) A plurality of word line trenches T2 extending in the first horizontal direction (X direction) may be in the substrate **110**, and a plurality of gate dielectric layers **116**, a plurality of word lines **118**, and a buried insulating layer **120** may be in the plurality of word line trenches T2. The plurality of word lines **118** may correspond to the plurality of word lines WL illustrated in FIG. 1.

(18) The gate dielectric layer **116** may include at least one of a silicon oxide layer, a silicon nitride layer, a silicon oxynitride layer, an oxide/nitride/oxide (ONO) layer, and a high-k dielectric layer having a higher dielectric constant than the silicon oxide layer. The high-k dielectric layer may include HfO₂, Al₂O₃, HfAlO₃, Ta₂O₃, TiO₂, or a combination thereof. The plurality of word lines **118** may include Ti, TiN, Ta, TaN, W, WN, TiSiN, WSiN, or a

combination thereof. The plurality of buried insulating layers **120** may include a silicon oxide layer, a silicon nitride layer, a silicon oxynitride layer, or a combination thereof.

(19) A buffer layer **122** may be on the substrate **110**. The buffer layer **122** may cover top surfaces of the plurality of active regions ACT, a top surface of the device isolation layer **112**, and top surfaces of the plurality of buried insulating layers **120**. According to an example embodiment, the buffer layer **122** may include a first silicon oxide layer, a silicon nitride layer, and a second silicon oxide layer sequentially formed on the substrate **110**. However, the disclosure is not limited thereto, and the buffer layer **122** may include a different arrangement.

(20) A plurality of bit lines BL extending parallel to each other in the second horizontal direction (Y direction) may be on the buffer layer **122**. The plurality of bit lines BL may be apart from each other in the first horizontal direction (X direction). A direct contact DC may be on a portion of each of the plurality of active regions ACT. Each of the plurality of bit lines BL may be connected to the active region ACT via the direct contact DC. The direct contact DC may include Si, Ge, W, WN, Co, Ni, Al, Mo, Ru, Ti, TiN, Ta, TaN, Cu, or a combination thereof. According to an example embodiment, the direct contact DC may include a doped polysilicon layer. According to an example embodiment, a plurality of recess spaces R1 that expose the active region ACT of the substrate **110** between each of the plurality of bit lines BL may be formed.

(21) Each of the plurality of bit lines BL may include a lower conductive layer **130**, an intermediate conductive layer **132**, and an upper conductive layer **134** sequentially stacked on the substrate **110**. Each of the plurality of bit lines BL may be covered by an insulating capping pattern **136**. In a vertical direction (Z direction), the insulating capping pattern **136** may be on the upper conductive layer **134**. A top surface of the lower conductive layer **130** of the bit line BL and a top surface of the direct contact DC may be on the same plane. In FIGS. 2A-2C, it is illustrated that the plurality of bit lines BL have a triple conductive layer structure including the lower conductive layer **130**, the intermediate conductive layer **132**, and the upper conductive layer **134**, but the inventive concept is not limited thereto. For example, according to another example embodiment, the plurality of bit lines BL may have a single conductive layer, a double conductive layer, or a stack structure of a plurality of conductive layers of four or more conductive layers.

(22) According to an example embodiment, the bottom conductive layer **130** may include a doped polysilicon layer. Each of the intermediate conductive layer **132** and the upper conductive layer **134** may include a layer including Ti, TiN, TiSiN, tungsten (W), WN, tungsten silicide (WSix), tungsten silicon nitride (WSixNy), ruthenium (Ru), or combination thereof. For example, the intermediate conductive layer **132** may include a TiN layer and/or a TiSiN layer, and the upper conductive layer **134** may include a layer including Ti, TiN, W, WN, WSixNy, Ru, or a combination thereof. The insulating capping pattern **136** may include a silicon nitride layer.

(23) A plurality of contact plugs **150** may be on the substrate **110**. The plurality of contact plugs **150** may have a pillar shape extending in the vertical direction (Z direction) in a space between each of the plurality of bit lines BL. Each of the plurality of contact plugs **150** may contact the active region ACT. The lower end of each of the plurality of contact plugs **150** may be at a level lower than the top surface of the substrate **110** so that the lower edge of each of the plurality of contact plugs **150** is buried in the substrate **110**. The plurality of contact plugs **150** may include a semiconductor material doped with impurities, a metal, a conductive metal nitride, or a combination thereof, but is not limited thereto.

(24) In the integrated circuit device **100**, one direct contact DC and a pair of contact plugs **150** facing each other with the one direct contact DC therebetween may be connected to different active regions AC among the plurality of active regions AC.

(25) The plurality of contact plugs **150** may be arranged in a row in the second horizontal direction (Y direction) between a pair of bit lines BL adjacent to each other selected from the plurality of bit lines BL. An insulating fence **149** may be between each of the plurality of contact plugs **150** arranged in a row in the second horizontal direction (Y direction). The plurality of contact plugs

150 may be insulated from each other by a plurality of insulating fences **149**. Each of the plurality of insulating fences **149** may have a pillar shape extending in the vertical direction (Z direction) on the substrate **110**. According to an example embodiment, the plurality of insulating fences **149** may include a silicon nitride layer.

(26) The integrated circuit device **100** may include a plurality of spacer structures **SP1** between the plurality of bit lines **BL** and the plurality of contact plugs **150**. One spacer structure **SP1** may be between one bit line **BL** and the plurality of contact plugs **150** arranged in a row in the second horizontal direction (Y direction). Each of the plurality of spacer structures **SP1** may include an inner oxide layer **140**, a carbon-containing oxide layer **142**, a gap fill insulating pattern **144**, an intermediate insulating spacer **146**, and an outer insulating spacer **148**.

(27) The inner oxide layers **140** may respectively contact a sidewall of the direct contact **DC** and a sidewall of the lower conductive layer **130** of the bit line **BL**. According to an example embodiment, the inner oxide layers **140** may directly contact the sidewall of the direct contact **DC** and the sidewall of the lower conductive layer **130** of the bit line **BL**. According to an example embodiment, the inner oxide layer **140** may be apart from the contact plug **150** with the carbon-containing oxide layer **142** therebetween. The inner oxide layer **140** may not include a portion contacting the contact plug **150**.

(28) In the vertical direction (Z direction), the uppermost surface of the inner oxide layer **140** may be at the same level as the uppermost surface of the lower conductive layer **130** of the bit line **BL** and the uppermost surface of the direct contact **DC**. Both sidewalls of the lower conductive layer **130** of the bit line **BL** may be covered with the inner oxide layer **140** from the lowermost level to the uppermost level in the vertical direction (Z direction). In addition, both sidewalls of the direct contact **DC** may be covered with the inner oxide layer **140** from the lowermost level to the uppermost level in the vertical direction (Z direction).

(29) The inner oxide layer **140** may include a silicon oxide layer. For example, when each of the lower conductive layer **130** of the bit line **BL** and the direct contact **DC** includes a doped polysilicon layer and each of the lower conductive layer **130** and the direct contact **DC** contact a silicon nitride layer instead of the silicon oxide layer, a depletion region near interfaces of the silicon nitride layer between the lower conductive layer **130** and the direct contact **DC** may be formed, and thus, electrical characteristics of the direct contact **DC** and the lower conductive layer **130** may be deteriorated. According to a technical aspect of the inventive concept, when the lower conductive layer **130** includes a doped polysilicon layer, by forming the inner oxide layers respectively contacting the sidewall of the lower conductive layer **130** with a silicon oxide layer, formation of the depletion region in the lower conductive layer **130** near the interface between the lower conductive layer **130** and the inner oxide layer **140** may be prevented. Similarly, when the direct contact **DC** includes a doped polysilicon layer, by forming the inner oxide layers respectively contacting the sidewall of the direct contact **DC** with a silicon oxide layer, formation of the depletion region in the direct contact **DC** near the interface between the direct contact **DC** and the inner oxide layer **140** may be prevented. Accordingly, deterioration of the electrical characteristics of the lower conductive layer **130** and the direct contact **DC** may be prevented.

(30) The carbon-containing oxide layer **142** may contact sidewalls of the intermediate conductive layer **132** and the upper conductive layer **134** of the bit line **BL**, and a sidewall of the insulating capping pattern **136**. The carbon-containing oxide layer **142** may nonlinearly extend on the sidewall of the bit line **BL** in the vertical direction (Z direction).

(31) The carbon-containing oxide layer **142** may include at least one protrusion **142PR**. The at least one protrusion **142PR** may have a shape protruding outward from the bit line **BL**, which is adjacent thereto. In other words, the protrusion **142PR** of the carbon-containing oxide layer **142** may have a structure that protrudes in a direction away from the bit line **BL** adjacent to the carbon-containing oxide layer **142**. The at least one protrusion **142PR** may include the protrusion **142PR** arranged at a level higher than an upper surface of the gap fill insulating pattern **144** in the vertical direction (Z

direction).

(32) The carbon-containing oxide layer **142** may include a portion contacting the direct contact DC. In addition, the carbon-containing oxide layer **142** may include a first portion between the bit line BL and the intermediate insulating spacer **146**, a second portion contacting the sidewall of the inner oxide layer **140**, and a third portion contacting a bottom surface of the intermediate insulating spacer **146**. The first portion, the second portion, and the third portion may be integrally connected to each other. The at least one protrusion **142PR** may include a protrusion **142PR** including the third portion.

(33) The carbon-containing oxide layer **142** may include a material having a lower dielectric constant than that of the silicon oxide layer. According to an example embodiment, the carbon-containing oxide layer **142** may include an SiOC layer. Carbon (C) content in the SiOC layer constituting the carbon-containing oxide layer **142** may be about 10 atomic % to about 50 atomic %. For example, the carbon-containing oxide layer **142** may be expressed as $\text{Si.sub.xO.sub.yC.sub.z}$, wherein $0.1 \leq x \leq 0.5$, $0.1 \leq y \leq 0.5$, and $0.1 \leq z \leq 0.8$, but are not limited thereto.

(34) According to an example embodiment, a portion of the carbon-containing oxide layer **142** contacting the bit line BL and the insulating capping pattern **136** may substantially have a constant thickness in the vertical direction (Z direction). In the first horizontal direction (X direction), the carbon-containing oxide layer **142** may have a thickness of about 10 Å (angstrom) to about 30 Å (angstrom).

(35) In the integrated circuit device **100**, because the sidewall of the bit line BL is covered by the carbon-containing oxide film **142** having a relatively low dielectric constant, undesired parasitic capacitance between the bit line BL and conductive regions adjacent thereto, for example, the contact plug **150**, may be reduced.

(36) The gap fill insulating pattern **144** may be between a lower edge of the contact plug **150** and the direct contact DC, and may cover the lower edge sidewall of the contact plug **150** and the sidewall of the direct contact DC. The sidewall and bottom surface of the gap fill insulating pattern **144** may be surrounded by the carbon-containing oxide layer **142**. Each of the inner oxide layer **140** and the carbon-containing oxide layer **142** may include a portion between the direct contact DC and the gap fill insulating pattern **144**.

(37) The inner oxide layer **140** may be apart from the contact plug **150** with the carbon-containing oxide layer **142** and the gap fill insulating pattern **144** therebetween.

(38) The intermediate insulating spacer **146** may cover sidewalls of the bit line BL, which is adjacent thereto. The intermediate insulating spacer **146** may be between the carbon-containing oxide layer **142** and the outer insulating spacer **148**. The intermediate insulating spacer **146** may include a silicon oxide layer, an air spacer, or a combination thereof. In the present specification, the term “air” may be referred to as the atmosphere or other gases that may be present during a manufacturing process.

(39) A portion of the inner oxide layer **140** may overlap a portion of the carbon-containing oxide layer **142** in the vertical direction (Z direction). The other portion of the inner oxide layer **140** may overlap the other portion of the carbon-containing oxide layer **142** in the first horizontal direction (X direction).

(40) The outer insulating spacer **148** may cover sidewalls of the bit line BL, which is adjacent thereto. The outer insulating spacer **148** may extend in the vertical direction (Z direction) to cover sidewalls of the bit line BL, which is adjacent thereto, with the carbon-containing oxide layer **142** and the intermediate insulating spacer **146** therebetween. The outer insulating spacer **148** may be apart from the carbon-containing oxide layer **142** with the intermediate insulating spacer **146** therebetween. According to an example embodiment, the outer insulating spacer **148** may include a silicon nitride layer.

(41) The at least one protrusion **142PR** of the carbon-containing oxide layer **142** may be between the adjacent bit line BL and the outer insulating spacer **148**, and may include the protrusion **142PR**

protruding toward the outer insulating spacer **148**.

(42) The inner oxide layer **140**, the carbon-containing oxide layer **142**, the intermediate insulating spacer **146**, and the outer insulating spacer **148** may each extend parallel to the bit line BL in a second horizontal direction (Y direction).

(43) A metal silicide layer **172** and a plurality of conductive landing pads LP may be sequentially formed on each of the plurality of contact plugs **150**. The plurality of conductive landing pads LP may be connected to the plurality of contact plugs **150** via the metal silicide layer **172**. The plurality of conductive landing pads LP may extend from the space between each of the plurality of insulating capping patterns **136** to the upper portion of each of the plurality of insulating capping patterns **136** so that the plurality of conductive landing pads LP vertically overlap a portion of the plurality of bit lines BL. Each of the plurality of conductive landing pads LP may include a conductive barrier layer **174** and a conductive layer **176**.

(44) According to an example embodiment, the metal silicide layer **172** may include cobalt silicide, nickel silicide, or manganese silicide, but is not limited thereto. According to an example embodiment, the metal silicide layer **172** may be omitted. The conductive barrier layer **174** may have a Ti/TiN stack structure. The conductive layer **176** may include doped polysilicon, metal, metal silicide, conductive metal nitride, or a combination thereof. For example, the conductive layer **176** may include tungsten (W). The plurality of conductive landing pads LP may have a plurality of an island-type pattern shapes in a plan view. The plurality of conductive landing pads LP may be electrically insulated from each other by an insulating layer **180** filling a space therearound. According to an example embodiment, the plurality of conductive landing pads LP may be insulated from each other by filling an upper recess space R2 around the plurality of conductive landing pads LP with the insulating layer **180**.

(45) FIG. 3 is a cross-sectional view of an integrated circuit device **100A** according to an embodiment of the inventive concept. In FIG. 3, some components of a portion, of the integrated circuit device **100A**, corresponding to the dashed region AX in (a) of FIG. 2 are enlarged.

(46) Referring to FIG. 3, the integrated circuit device **100A** may have substantially the same configuration as the integrated circuit device **100** described with reference to FIGS. 2A-2C, particularly the AX region illustrated in FIG. 2C. However, the integrated circuit device **100A** may include a plurality of spacer structures SP1A instead of the plurality of spacer structures SP1.

(47) The plurality of spacer structures SP1A may have substantially the same structure as the spacer structures SP1 illustrated in FIGS. 2A-2C. However, the plurality of spacer structures SP1A may include a carbon-containing oxide layer **142A** and an intermediate insulating spacer **146A**, which have variable thicknesses in the vertical direction (Z direction). According to an example embodiment, the carbon-containing oxide layer **142A** may have a non-uniform thicknesses in the vertical direction (Z direction). According to an example embodiment, the intermediate insulating spacer **146A** may have a non-uniform thicknesses in the vertical direction (Z direction).

(48) The carbon-containing oxide layer **142A** may have substantially the same configuration as that described for the carbon-containing oxide layer **142** with reference to FIGS. 2A-2C. However, a first portion of the carbon-containing oxide layer **142A** contacting the bit line BL and a second portion of the carbon-containing oxide layer **142A** contacting the insulating capping pattern **136** may have different thicknesses in the vertical direction (Z direction). In the first horizontal direction (X direction), a first thickness W11 of the portion of the carbon-containing oxide layer **142A** contacting the upper conductive layer **134** of the bit line BL may be less than a second thickness W12 of a portion contacting the insulating capping pattern **136**. This may be due to a difference between deposition characteristics on a surface of the upper conductive layer **134** and deposition characteristics on a surface of the insulating capping pattern **136** in the process for forming the carbon-containing oxide layer **142A**, and a difference between reactions of the upper conductive layer **134** and the insulating capping pattern **136** with respect to pre-processing conditions in the pre-processing operation of forming the carbon-containing oxide layer **142**.

(49) According to an example embodiment, the carbon-containing oxide layer **142A** may have a thickness of about 10 Å to about 30 Å in the first horizontal direction (X direction). The difference between the second thickness **W12** and the first thickness **W11** of the carbon-containing oxide layer **142A** may be about 0.1 Å to about 20 Å, but is not limited thereto.

(50) The intermediate insulating spacer **146A** may have substantially the same configuration as that described for the intermediate insulating spacer **146** with reference to FIGS. 2A-2C. However, a width of a first portion of the intermediate insulating spacer **146A** facing the upper conductive layer **134** in the first horizontal direction (X direction) may be greater than a width of a second portion of the intermediate insulating spacer **146A**. According to an example embodiment, the second portion of the intermediate insulating spacer **146A** does not face the upper conductive layer **134**.

(51) FIG. 4 is a cross-sectional view of an integrated circuit device **100B** according to another example embodiment of the inventive concept. In FIG. 4, some components of a portion, of the integrated circuit device **100B**, corresponding to the dashed region AX in FIG. 2A are enlarged.

(52) Referring to FIG. 4, the integrated circuit device **100B** may have substantially the same configuration as the integrated circuit device **100** described with reference to FIGS. 2A-2C. However, the integrated circuit device **100B** may include a plurality of spacer structures **SP1B** instead of the plurality of spacer structures **SP1**. The plurality of spacer structures **SP1B** may have substantially the same structure as the spacer structures **SP1** illustrated in FIGS. 2A-2C. However, the plurality of spacer structures **SP1B** may include an intermediate insulating spacer **146B** including an air spacer **AS1** and an intermediate insulating spacer pattern **P1**, instead of the intermediate insulating spacer **146**. The intermediate insulating spacer pattern **P1** among the air spacer **AS1** and the intermediate insulating spacer pattern **P1**, which constitute the intermediate insulating spacer **146B**, may be closer to the substrate **110** (refer to FIGS. 2A-2C).

(53) In the integrated circuit device **100B**, because the sidewall of the bit line BL is covered with the intermediate insulating spacer **146B** including the air spacer **AS1** having a relatively low dielectric constant, undesired parasitic capacitance between the bit line BL and the conductive regions adjacent thereto, for example, the contact plug **150**, may be reduced.

(54) FIGS. 5A-5C are cross-sectional views of an integrated circuit device **200** according to another example embodiment of the inventive concept. FIG. 5A is a cross-sectional view of some components of a portion corresponding to a cross-section taken along line A-A' of FIG. 1, FIG. 5B is a cross-sectional view of some components of a portion corresponding to a cross-section taken along line B-B' of FIG. 1, and FIG. 5C is an enlarged cross-sectional view of a portion corresponding to a dash-lined region AX in FIG. 5A. In FIGS. 5A-5C, the same reference numerals as those in FIGS. 1 and FIGS. 2A-2C denote the same members, and detailed descriptions thereof are omitted.

(55) Referring to FIGS. 5A-5C, the integrated circuit device **200** may have substantially the same configuration as the integrated circuit device **100** described with reference to FIGS. 2A-2C. However, the integrated circuit device **200** may include a plurality of spacer structures **SP2** instead of the plurality of spacer structures **SP1**.

(56) Each of the plurality of spacer structures **SP2** may include an inner oxide layer **240**, a carbon-containing oxide layer **242**, a gap fill insulating pattern **244**, an intermediate insulating spacer **246**, and the outer insulating spacer **148**.

(57) The inner oxide layers **240** may respectively contact a sidewall of the direct contact DC and a sidewall of the lower conductive layer **130** of the bit line BL. The inner oxide layer **240** may include a portion between the direct contact DC and a gap fill insulating pattern **244**. The inner oxide layer **240** may include a portion contacting the contact plug **150**.

(58) The inner oxide layer **240** may include a silicon oxide layer. When each of the direct contact DC and the lower conductive layer **130** of the bit line BL is a doped polysilicon layer, by forming the inner oxide layer **140** contacting each of the sidewall of the lower conductive layer **130** and the sidewall of the direct contact DC with a silicon oxide layer, formation of a depletion region near

the interface of the inner oxide layer **140** in each of the lower conductive layer **130** and the direct contact DC may be prevented, and accordingly, deterioration of electrical characteristics of the lower conductive layer **130** and the direct contact DC may be prevented.

(59) The carbon-containing oxide layer **242** may be apart from the contact plug **150** closest to the carbon-containing oxide layer **242** with the outer insulating spacer **148** therebetween. The carbon-containing oxide layer **242** may not include a portion between the direct contact DC and the gap fill insulating pattern **244**.

(60) The gap fill insulating pattern **244** may be between the lower edge of the contact plug **150** and the direct contact DC. The gap fill insulating pattern **244** may contact the direct contact DC and the inner oxide layer **240**. In the first horizontal direction (X direction), the gap fill insulating pattern **244** may face the direct contact DC with the inner oxide layer **240** therebetween. The outer insulating spacer **148** may cover the sidewall of the bit line BL and the sidewall of the insulating capping pattern **136** on the gap fill insulating pattern **244**.

(61) The carbon-containing oxide layer **242** may contact sidewalls of the intermediate conductive layer **132** and the upper conductive layer **134** of the bit line BL, and a sidewall of the insulating capping pattern **136**. The carbon-containing oxide layer **242** may nonlinearly extend on the sidewall of the bit line BL in the vertical direction (Z direction). The carbon-containing oxide layer **242** may include a protrusion **242PR** protruding from the adjacent bit line BL toward the outer insulating spacer **148**. The protrusion **242PR** may be at a higher level than the upper surface of the gap fill insulating pattern **144** in the vertical direction (Z direction).

(62) The outer insulating spacer **148** may be between the carbon-containing oxide layer **242** and the contact plug **150**. The carbon-containing oxide layer **242** may not include a portion contacting the direct contact DC.

(63) The carbon-containing oxide layer **242** may include a portion between the bit line BL and the intermediate insulating spacer **146**, and a portion contacting the bottom surface of the intermediate insulating spacer **146**. A portion of the carbon-containing oxide layer **242** contacting the bottom surface of the intermediate insulating spacer **146** may constitute a protrusion **242PR**.

(64) According to an example embodiment, a portion of the carbon-containing oxide layer **242** contacting the bit line BL and the insulating capping pattern **136** may substantially have a constant thickness in the vertical direction (Z direction).

(65) In the integrated circuit device **200**, because the sidewall of the bit line BL is covered by the carbon-containing oxide layer **242** having a relatively low dielectric constant, undesired parasitic capacitance between the bit line BL and conductive regions adjacent thereto, for example, the contact plug **150**, may be reduced.

(66) The intermediate insulating spacer **246** may cover sidewalls of the bit line BL, which is adjacent thereto. The intermediate insulating spacer **246** may be between the carbon-containing oxide layer **242** and the outer insulating spacer **148**. The intermediate insulating spacer **246** may include a silicon oxide layer, an air spacer, or a combination thereof.

(67) A portion of the inner oxide layer **240** may overlap a portion of the carbon-containing oxide layer **242** in the vertical direction (Z direction). The other portion of the inner oxide layer **240** may overlap the other portion of the carbon-containing oxide layer **242** in the first horizontal direction (X direction).

(68) The outer insulating spacer **148** may be apart from the carbon-containing oxide layer **242** with the intermediate insulating spacer **246** therebetween.

(69) The inner oxide layer **240**, the carbon-containing oxide layer **122**, the intermediate insulating spacer **246**, and the outer insulating spacer **148** may each extend parallel to the bit line BL in the second horizontal direction (Y direction).

(70) More detailed configuration of the inner oxide layer **240**, the carbon-containing oxide layer **242**, the gap fill insulating pattern **244**, and the intermediate insulating spacer **246** may be substantially the same as descriptions of the inner oxide layer **140**, the carbon-containing oxide

layer **142**, the gap fill insulating pattern **144**, and the intermediate insulating spacer **146** given with reference to FIGS. **2A-2C**.

(71) FIG. **6** is a cross-sectional view of an integrated circuit device **200A** according to another embodiment of the inventive concept. In FIG. **6**, some components of a portion, of the integrated circuit device **200A**, corresponding to the dashed region AX in (a) of FIG. **5** are enlarged.

(72) Referring to FIG. **6**, the integrated circuit device **200A** may have substantially the same configuration as the integrated circuit device **200** described with reference to FIGS. **5A-5C**. However, the integrated circuit device **200A** may include a plurality of spacer structures SP2A instead of the plurality of spacer structures SP2.

(73) The plurality of spacer structures SP2A may have substantially the same structure as the spacer structures SP2 illustrated in FIGS. **5A-5C**. However, the plurality of spacer structures SP2A may include a carbon-containing oxide layer **242A** and an intermediate insulating spacer **246A**, which have variable thicknesses in the vertical direction (Z direction).

(74) The carbon-containing oxide layer **242A** may have substantially the same configuration as that described for the carbon-containing oxide layer **242** with reference to FIGS. **5A-5C**. However, a portion of the carbon-containing oxide layer **242A** contacting the bit line BL and a portion contacting the insulating capping pattern **136** may have different thicknesses in the vertical direction (Z direction). In the first horizontal direction (X direction), a first thickness W21 of the portion of the carbon-containing oxide layer **242A** contacting the upper conductive layer **134** of the bit line BL may be less than a second thickness W22 of a portion contacting the insulating capping pattern **136**. The carbon-containing oxide layer **242A** may have a thickness of about 10 Å to about 30 Å. The difference between the second thickness W22 and the first thickness W21 of the carbon-containing oxide layer **242A** may be about 0.1 Å to about 20 Å, but is not limited thereto.

(75) The intermediate insulating spacer **246A** may have substantially the same configuration as that described for the intermediate insulating spacer **246** with reference to FIGS. **5A-5C**. However, a width of a portion of the intermediate insulating spacer **246A** facing the upper conductive layer **134** in the first horizontal direction (X direction) may be greater than a width of the other portion of the intermediate insulating spacer **246A**.

(76) FIG. **7** is a cross-sectional view of an integrated circuit device **200B** according to another example embodiment of the inventive concept. In FIG. **7**, some components of a portion, of the integrated circuit device **200B**, corresponding to the dashed region AX in FIG. **5A** are enlarged.

(77) Referring to FIG. **7**, the integrated circuit device **200B** may have substantially the same configuration as the integrated circuit device **200** described with reference to FIGS. **5A-5C**. However, the integrated circuit device **200B** may include a plurality of spacer structures SP2B instead of the plurality of spacer structures SP2. The plurality of spacer structures SP2B may have substantially the same structure as the spacer structures SP2 illustrated in FIGS. **5A-5C**. However, the plurality of spacer structures SP2B may include an intermediate insulating spacer **246B** including an air spacer AS2 and an intermediate insulating spacer pattern P2, instead of the intermediate insulating spacer **246**. The intermediate insulating spacer pattern P2 among the air spacer AS2 and the intermediate insulating spacer pattern P2, which constitute the intermediate insulating spacer **246B**, may be closer to the substrate **110** (refer to FIGS. **5A-5C**).

(78) In the integrated circuit device **200B**, because the sidewall of the bit line BL is covered with the intermediate insulating spacer **246B** including the air spacer AS2 having a relatively low dielectric constant, undesired parasitic capacitance between the bit line BL and the conductive regions adjacent thereto, for example, the contact plug **150**, may be reduced.

(79) FIG. **8** is a cross-sectional view of an integrated circuit device **300** according to another example embodiment of the inventive concept. In FIG. **8**, some components of a portion, of the integrated circuit device **300**, corresponding to the dashed region AX in FIG. **5A** are enlarged.

(80) Referring to FIG. **8**, the integrated circuit device **300** may have substantially the same configuration as the integrated circuit device **200** described with reference to FIGS. **5A-5C**.

However, the integrated circuit device **300** may include a plurality of spacer structures **SP3** instead of the plurality of spacer structures **SP2**.

(81) The plurality of spacer structures **SP3** may have substantially the same structure as the spacer structures **SP2** illustrated in FIGS. 5A-5C. However, the plurality of spacer structures **SP3** may include an inner oxide layer **340**, a carbon-containing oxide layer **342**, a gap fill insulating pattern **344**, an intermediate insulating spacer **346**, and the outer insulating spacer **148**.

(82) The inner oxide layer **340** may have substantially the same configuration as the inner oxide layer **240** described with reference to FIGS. 5A-5C. However, an uppermost level of the inner oxide layer **340** may be lower than the uppermost level of the direct contact **DC**. The sidewall of a portion of the upper side of the direct contact **DC** and the sidewall of the lower conductive layer **130** of the bit line **BL** may not be covered with the inner oxide layer **340**.

(83) The carbon-containing oxide layer **342** may have substantially the same configuration as the carbon-containing oxide layer **242** described with reference to FIGS. 5A-5C. However, the carbon-containing oxide layer **342** may contact a sidewall of a portion of the upper side of the direct contact **DC** and a sidewall of the lower conductive layer **130** of the bit line **BL**.

(84) The gap fill insulating pattern **344** may be between the lower edge of the contact plug **150** and a portion of the bottom surface of the direct contact **DC**.

(85) The carbon-containing oxide layer **342** may contact the sidewall of each of the lower conductive layer **130**, the intermediate conductive layer **132** and the upper conductive layer **134** of the bit line **BL**, and the sidewall of the insulating capping pattern **136**. The carbon-containing oxide layer **342** may nonlinearly extend on the sidewall of the bit line **BL** in the vertical direction (**Z** direction). The carbon-containing oxide layer **342** may include a protrusion **342PR** protruding from each of the lower conductive layer **130** and the direct contact **DC** of the adjacent bit line **BL** toward the outer insulating spacer **148**. The protrusion **342PR** may be at a higher level than an upper surface of a gap fill insulating pattern **344** in the vertical direction (**Z** direction). The outer insulating spacer **148** may be between the carbon-containing oxide layer **342** and the contact plug **150**. The carbon-containing oxide layer **342** may not include a portion contacting the direct contact **DC**.

(86) The carbon-containing oxide layer **342** may include a portion between the bit line **BL** and the intermediate insulating spacer **346**, and a portion contacting the bottom surface of the intermediate insulating spacer **346**. A portion of the carbon-containing oxide layer **342** contacting the bottom surface of the intermediate insulating spacer **346** may constitute a protrusion **342PR**.

(87) According to an example embodiment, a portion of the carbon-containing oxide layer **342** contacting the bit line **BL** and the insulating capping pattern **136** may substantially have a constant thickness in the vertical direction (**Z** direction).

(88) In the integrated circuit device **300**, because the sidewall of the bit line **BL** is covered by the carbon-containing oxide layer **342** having a relatively low dielectric constant, undesired parasitic capacitance between the bit line **BL** and conductive regions adjacent thereto, for example, the contact plug **150**, may be reduced.

(89) The intermediate insulating spacer **346** may cover sidewalls of the bit line **BL**, which is adjacent thereto. The intermediate insulating spacer **346** may be between the carbon-containing oxide layer **342** and the outer insulating spacer **148**. The intermediate insulating spacer **346** may include a silicon oxide layer, an air spacer, or a combination thereof.

(90) A portion of the inner oxide layer **340** may overlap a portion of the carbon-containing oxide layer **342** in the vertical direction (**Z** direction).

(91) More detailed configuration of the inner oxide layer **340**, the carbon-containing oxide layer **342**, the gap fill insulating pattern **344**, and the intermediate insulating spacer **346** may be substantially the same as descriptions of the inner oxide layer **140**, the carbon-containing oxide layer **142**, the gap fill insulating pattern **144**, and the intermediate insulating spacer **146** given with reference to FIGS. 2A-2C.

(92) FIG. 9 is a cross-sectional view of an integrated circuit device **300A** according to another embodiment of the inventive concept. In FIG. 9, some components of a portion, of the integrated circuit device **300A**, corresponding to the dashed region AX in FIG. 5A are enlarged.

(93) Referring to FIG. 9, the integrated circuit device **300A** may have substantially the same configuration as the integrated circuit device **300** described with reference to FIG. 8. However, the integrated circuit device **300A** may include a plurality of spacer structures SP3A instead of the plurality of spacer structures SP3.

(94) The plurality of spacer structures SP3A may have substantially the same structure as the spacer structures SP3 illustrated in FIG. 8. However, the plurality of spacer structures SP3A may include a carbon-containing oxide layer **342A** and an intermediate insulating spacer **346A**, which have variable thicknesses in the vertical direction (Z direction).

(95) The carbon-containing oxide layer **342A** may have substantially the same configuration as that described for the carbon-containing oxide layer **342** with reference to FIG. 8. However, a portion of the carbon-containing oxide layer **342A** contacting the bit line BL and a portion contacting the insulating capping pattern **136** may have different thicknesses in the vertical direction (Z direction). In the first horizontal direction (X direction), a first thickness W31 of the portion of the carbon-containing oxide layer **342A** contacting the upper conductive layer **134** of the bit line BL may be less than a second thickness W32 of a portion of the carbon-containing oxide layer **342A** contacting the insulating capping pattern **136**. The carbon-containing oxide layer **342A** may have a thickness of about 10 Å to about 30 Å. The difference between the second thickness W32 and the first thickness W31 of the carbon-containing oxide layer **342A** may be about 0.1 Å to about 20 Å, but is not limited thereto.

(96) The intermediate insulating spacer **346A** may have substantially the same configuration as that described for the intermediate insulating spacer **346** with reference to FIG. 8. However, a width of a portion of the intermediate insulating spacer **346A** facing the upper conductive layer **134** in the first horizontal direction (X direction) may be greater than a width of the other portion of the intermediate insulating spacer **346A**.

(97) FIG. 10 is a cross-sectional view of an integrated circuit device **300B** according to another example embodiment of the inventive concept. In FIG. 10, some components of a portion, of the integrated circuit device **300B**, corresponding to the dashed region AX in FIG. 5A are enlarged.

(98) Referring to FIG. 10, the integrated circuit device **300B** may have substantially the same configuration as the integrated circuit device **300** described with reference to FIG. 8. However, the integrated circuit device **300B** may include a plurality of spacer structures SP3B instead of the plurality of spacer structures SP3. The plurality of spacer structures SP3B may have substantially the same structure as the spacer structures SP3 illustrated in FIG. 8. However, the plurality of spacer structures SP3B may include an intermediate insulating spacer **346B** including an air spacer AS3 and an intermediate insulating spacer pattern P3, instead of the intermediate insulating spacer **346**.

(99) In the integrated circuit device **300B**, because the sidewall of the bit line BL is covered with the intermediate insulating spacer **346B** including the air spacer AS3 having a relatively low dielectric constant, undesired parasitic capacitance between the bit line BL and the conductive regions adjacent thereto, for example, the contact plug **150**, may be reduced.

(100) FIGS. 11A through 11O are cross-sectional views illustrating a method of manufacturing an integrated circuit device according to process sequences, according to the technical idea of the inventive concept. In FIGS. 11A through 11O, (a) is a cross-sectional view of some components according to a process sequence of some region corresponding to a cross-section taken along line A-A' in FIG. 1, and (b) is a cross-sectional view of some components according to a process sequence of some region corresponding to a cross-section taken along line B-B' in FIG. 1. In FIGS. 11G through 11O, (c) is an enlarged cross-sectional view of a portion corresponding to the dashed region AX in (a) of the corresponding figure. A manufacturing method of the integrated

circuit device **100** illustrated in FIGS. 2A, 2B and 2C is described with reference to FIGS. 11A through 11O.

(101) Referring to FIG. 11A, the device isolation trench T1 may be formed in the substrate **110**, and the device isolation layer **112** may be formed in the device isolation trench T1. The plurality of active regions ACT may be defined in the substrate **110** by the element isolation layer **112**.

(102) The plurality of word line trenches T2 may be formed in the substrate **110**. The plurality of word line trenches T2 may extend parallel to each other in the first horizontal direction (X direction), and may have a line shape crossing the active region ACT. According to an example embodiment, in order to form the plurality of word line trenches T2 including steps on the bottom surface thereof, each of the device isolation layer **112** and the substrate **110** may be etched by a separate etching process, and an etching depth of the device isolation layer **112** may be manufactured different from an etching depth of the substrate **110**. After cleaning the result of forming the plurality of word line trenches T2, the gate dielectric layer **116**, the word line **118**, and the buried insulating layer **120** may be sequentially formed in each of the plurality of word line trenches T2. Before or after forming the plurality of word lines **118**, an ion implantation process for forming a plurality of source/drain regions on the plurality of active regions ACT may be performed.

(103) The buffer layer **122** may be formed on the substrate **110**. The buffer layer **122** may cover top surfaces of the plurality of active regions ACT, a top surface of the device isolation layer **112**, and top surfaces of the plurality of buried insulating layers **120**. The buffer layer **122** may include a first silicon oxide layer, a silicon nitride layer, and a second silicon oxide layer sequentially formed on the substrate **110**, but is not limited thereto.

(104) Referring to FIG. 11B, the lower conductive layer **130** may be formed on the buffer layer **122**. The lower conductive layer **130** may include a doped polysilicon layer.

(105) Referring to FIG. 11C, after forming the mask pattern MP1 on the lower conductive layer **130**, a direct contact hole DCH exposing the active region ACT of the substrate **110** may be formed by etching a portion of each of the lower conductive layer **130** exposed through an opening MH of the mask pattern MP1, the buffer layer **122** below the mask pattern MP1, and the device isolation layer **112**. The mask pattern MP1 may include an oxide layer, a nitride layer, or a combination thereof, but is not limited thereto.

(106) Referring to FIG. 11D, the mask pattern MP1 may be removed from the result illustrated in FIG. 11C, and the direct contact DC may be formed in a direct contact hole DCH.

(107) According to an example embodiment, in order to form the direct contact DC, a doped polysilicon layer having a thickness sufficient to fill the inside of the direct contact hole DCH and the direct contact hole DCH on the upper portion of the bottom conductive layer **130** may be formed, and then an unnecessary portion of the doped polysilicon layer may be removed so that the doped polysilicon layer remains only in the direct contact hole DCH.

(108) Referring to FIG. 11E, the intermediate conductive layer **132**, the upper conductive layer **134**, and the plurality of insulating capping patterns **136** may be sequentially formed on the lower conductive layer **130** and the direct contact DC. Each of the plurality of insulating capping patterns **136** may be formed of a line pattern extending long in the second horizontal direction (Y direction).

(109) Referring to FIG. 11F, a portion of each of the upper conductive layer **134**, the intermediate conductive layer **132**, the lower conductive layer **130**, and the direct contact DC may be etched by using the insulating capping pattern **136** as an etching mask to form the plurality of bit lines BL on the substrate **110**. The plurality of bit lines BL may include remaining portions of each of the lower conductive layer **130**, the intermediate conductive layer **132**, and the upper conductive layer **134**. After the plurality of bit lines BL are formed, a portion of the direct contact hole DCH may be exposed around the direct contact DC again, and a line space LS extending long in the second horizontal direction (Y direction) may be defined between each of the plurality of bit line structures each including the bit line BL and the insulating capping pattern **136**.

(110) Referring to FIG. 11G, the plurality of inner oxide layers **140** selectively covering the sidewall of each of the lower conductive layer **130** and the direct contact DC exposed in the resultant illustrated in FIG. 11F may be formed. The inner oxide layer **140** may contact both sidewalls of each of the lower conductive layer **130** and the direct contact DC. According to an example embodiment, a selective oxidation process may be performed on the exposed surfaces of each of the lower conductive layer **130** and the direct contact DC to form the plurality of inner oxide layers **140**.

(111) Referring to FIG. 11H, the carbon-containing oxide layer **142** covering the surface exposed after the formation of the inner oxide layers **140** illustrated in FIG. 11G may be formed. For instance, the carbon-containing oxide layer **142** may be formed to conform to the exposed layers, so as to cover the exposed layers in FIG. 11G. The carbon-containing oxide layer **142** may contact each of the inner oxide layer **140**, the intermediate conductive layer **132**, the upper conductive layer **134**, and the plurality of insulating capping patterns **136**.

(112) A chemical vapor deposition (CVD) process or an atomic layer deposition (ALD) process may be used to form the carbon-containing oxide layer **142**.

(113) According to an example embodiment, the carbon-containing oxide layer **142** may substantially have a constant thickness in the vertical direction (Z direction) on the sidewall of the bit line BL and the sidewall of the insulating capping pattern **136**. In other exemplary embodiments, by using the difference between deposition characteristics on the surface of the upper conductive layer **134** and deposition characteristics on the surface of the insulating capping pattern **136**, the difference between reactions of the upper conductive layer **134** and the insulating capping pattern **136** with respect to pre-processing conditions in the pre-processing operation of forming the carbon-containing oxide layer **142** or the like, the carbon-containing layer **142A** illustrated in FIG. 3, instead of the carbon-containing oxide layer **142**, may be formed.

(114) Referring to FIG. 11I, a gap fill insulating layer P**144** covering sidewalls of each of the plurality of bit lines BL, the plurality of insulating capping patterns **136**, and the plurality of direct contacts DC while filling a remaining space of the direct contact hole DCH in the result illustrated in FIG. 11H may be formed.

(115) According to an example embodiment, the gap fill insulating layer P**144** may include a silicon nitride layer. A CVD or ALD process may be used to form the gap fill insulating layer P**144**.

(116) Referring to FIG. 11J, the gap fill insulating layer P**144** from the result illustrated in FIG. 11I may be isotropically etched to form the gap fill insulating pattern **144** including the remaining portion of the gap fill insulating layer P**144**. While the gap fill insulating layer P**144** is isotropically etched, the carbon-containing oxide layer **142** may function as an etch stop layer. The gap fill insulating pattern **144** may include a portion of the gap fill insulating layer P**144** that fills the inside of the direct contact hole DCH, and a portion covering an entrance of the direct contact hole DCH from the outside of the entrance of the direct contact hole DCH.

(117) Referring to FIG. 11K, after an intermediate insulating spacer layer conformally covering the surfaces exposed from the result illustrated in FIG. 11J by using a CVD or ALD process is formed, the plurality of intermediate insulating spacers **146** may be formed from the intermediate insulating spacer layer by isotropically etching the intermediate insulating spacer layer.

(118) While the intermediate insulating spacer layer is anisotropically etched to form the plurality of intermediate insulating spacers **146**, a portion of the buffer layer **122** and a portion of the carbon-containing oxide layer **142** covering the buffer layer **122** may be removed. As a result, a portion of the substrate **110**, a portion of the carbon-containing oxide layer **142**, and a portion of the gap fill insulating pattern **144** may be exposed at a bottom of a plurality of line spaces LS. Each of the plurality of intermediate insulating spacers **146** may cover the sidewall of the bit line BL and the sidewall of the insulating capping pattern **136** on the carbon-containing oxide layer **142**.

(119) The plurality of intermediate insulating spacers **146** may include a material different from the material of the carbon-containing oxide layer **142** and the material of the gap fill insulating pattern

144. The plurality of intermediate insulating spacers **146** may include a material having an etching selectivity with respect to the carbon-containing oxide layer **142** and the gap fill insulating pattern **144**. For example, the plurality of intermediate insulating spacers **146** may include a silicon oxide layer.

(120) Referring to FIG. **11L**, the outer insulating spacer **148** conformally covering the result illustrated in FIG. **11K** may be formed. The outer insulating spacer **148** may include a material having an etch selectivity with respect to the plurality of intermediate insulating spacers **146**. For example, the outer insulating spacer **148** may include a silicon nitride layer. A CVD or ALD process may be used to form the outer insulating spacer **148**.

(121) Referring to FIG. **11M**, the line space LS may be divided into a plurality of contact spaces CS by forming the plurality of insulating fences **149** apart from each other in the line space LS defined by the outer insulating spacer **148** between each of the bit lines BL from the result illustrated in FIG. **11L**.

(122) Each of the plurality of insulating fences **149** may vertically overlap the word line **118** on the word line **118**. The plurality of insulating fences **254** may include a silicon nitride layer. According to an example embodiment, while the plurality of insulating fences are formed, a portion of the plurality of insulating capping patterns **136** may be consumed, and a height of the plurality of insulating capping patterns **136** may be reduced.

(123) Next, by removing a portion of structures that are exposed via the plurality of contact spaces CS, the plurality of recess spaces R1 that expose the active region ACT of the substrate **110** between each of the plurality of bit lines BL may be formed. To form the plurality of recess spaces R1, an anisotropic etching process or a combination of an anisotropic etching process and an isotropic etching process may be used. For example, the plurality of recess spaces R1 may be formed by anisotropically etching a portion of the outer insulating spacer **148** exposed via the plurality of contact spaces CS between each of the plurality of bit lines BL and a portion of the substrate **110** under the outer insulating spacer **148**, and by isotropically etching a portion of the active region ACT of the substrate that is exposed as a result of the anisotropical etching. Each of the plurality of recess spaces R1 may communicate with the contact space CS. While the etching process for forming the contact space CS is performed, a portion of each of the inner insulating spacer **142** and the gap fill insulating pattern **144** may be consumed in a region adjacent to the top surface of the substrate **110**.

(124) A portion of the active region ACT of the substrate **110**, a portion of the carbon-containing oxide layer **142**, and a portion of the gap fill insulating pattern **144** may be exposed through the plurality of recess spaces R1.

(125) Referring to FIG. **11N**, the plurality of contact plugs **150** filling a portion of the contact space CS between each of the plurality of bit lines BL while filling the plurality of recess spaces R1 between each of a plurality of bit lines BL may be formed.

(126) Referring to FIG. **11O**, the metal silicide layer **172** and the plurality of conductive landing pads LP may be sequentially formed on the plurality of contact plugs **150** exposed through the plurality of contact spaces CS (refer to FIG. **11N**).

(127) The contact plug **150** and the metal silicide layer **172** may form at least a portion of the buried contact BC illustrated in FIG. **1**. The plurality of conductive landing pads LP may extend to the upper portion of the insulating capping pattern **136** while filling the plurality of contact spaces CS on the metal silicide layer **172** and vertically overlapping a portion of the plurality of bit lines BL. Each of the plurality of conductive landing pads LP may include the conductive barrier layer **174** and the conductive layer **176**.

(128) To form the plurality of conductive landing pads LP, after the conductive barrier layer **174** and the conductive layer **176** are formed on a front side of the result in which the metal silicide layer **172** has been formed, by forming a mask pattern (not illustrated) exposing a portion of the conductive layer **176** on the conductive layer **176**, and etching the conductive layer **176**, the

conductive barrier layer **174**, and insulating layers therearound by using the mask pattern as an etching mask, an upper recess space R2 may be formed. The mask pattern may include a silicon nitride layer, but is not limited thereto.

(129) The plurality of conductive landing pads LP may have a plurality of an island-type pattern shapes. Portions of the plurality of conductive landing pads LP extending in the horizontal direction from the outside of the contact space CS may constitute the plurality of conductive landing pads LP illustrated in FIG. 1.

(130) The plurality of conductive landing pads LP may be insulated from each other by filling the upper recess space R2 around the plurality of conductive landing pads LP with the insulating layer **180**. Next, a plurality of capacitor lower electrodes capable of being electrically connected to the plurality of conductive landing pads LP may be formed on the insulating layer **180**.

(131) According to an example embodiment, after the upper recess space R2 around the plurality of conductive landing pads LP is formed in the process described with reference to FIG. 11O, and before the upper recess space R2 is filled with the insulating layer **180**, at least a portion of the silicon oxide layer constituting the plurality of intermediate insulating spacers **146** may be removed through the upper recess space R2.

(132) In an example, the silicon oxide layer constituting the plurality of intermediate insulating spacers **146** may be completely removed through the upper recess space R2 so that the intermediate insulating spacer **146** is formed as an air spacer.

(133) According to another example embodiment, in order to manufacture the integrated circuit device **100B** illustrated in FIG. 4, after the upper recess space R2 is formed in the process described with reference to FIG. 11O, and before the upper recess space R2 is filled with the insulating layer **180**, the air spacer AS1 may be formed by removing a portion of the silicon oxide layer constituting the plurality of intermediate insulating spacer **146** through the upper recess space R2, and the intermediate insulating spacer pattern P1 including a remaining portion of the silicon oxide layer at the bottom portion of the air spacer AS1 may be maintained.

(134) FIGS. 12A through 12H are cross-sectional views illustrating a method of manufacturing an integrated circuit device according to another example embodiment of the inventive concept. In FIGS. 12A through 12H, (a) is a cross-sectional view of some components according to a process sequence of a portion corresponding to a cross-section taken along line A-A' of FIG. 1, (b) is a cross-sectional view of some components according to a process sequence of a portion corresponding to a cross-section taken along line B-B' of FIG. 1, and (c) is an enlarged cross-sectional view of a portion corresponding to a dash-lined region AX in (a). An example manufacturing method of the integrated circuit device **200** illustrated in FIGS. 5A-5C are described with reference to FIGS. 12A through 12H. In FIGS. 12A through 12H, the same reference numerals as those in FIGS. 1 through 11O may denote the same members, and descriptions thereof are omitted here.

(135) Referring to FIG. 12A, after the plurality of bit lines BL and the direct contacts DC are formed on the substrate **110** by performing the processes described with reference to FIGS. 11A through 11F, a preliminary inner oxide layer P240 covering the exposed surfaces after the formation of the plurality of bit lines BL and the direct contacts DC may be formed. The preliminary inner oxide layer P240 may include a silicon oxide layer. A CVD or ALD process may be used to form the preliminary inner oxide layer P240.

(136) Next, a gap fill insulating layer P244 may be formed on the preliminary inner oxide layer P240 in a similar manner to that described with reference to FIG. 11I. The remaining space of the direct contact hole DCH may be filled by the gap fill insulating layer P244 around the direct contact DC. The gap fill insulating layer P244 may include a silicon nitride layer. A thickness of the preliminary inner oxide layer P240 may be less than that of the gap fill insulating layer P244.

(137) Referring to FIG. 12B, in a method similar to that described with respect to the method of forming the gap fill insulating pattern **144** with reference to FIG. 11J, the gap fill insulating pattern

244 including the remaining portion of the gap fill insulating layer **P244** may be formed by isotropically etching the gap fill insulating layer **P244**.

(138) While the gap fill insulating layer **P244** is isotropically etched, the preliminary inner oxide layer **P240** may function as an etch stop layer protecting the bit line BL and the insulating capping pattern **136**. Portions of the preliminary inner oxide layer **P240** that are not covered by the gap fill insulating pattern **244** may be removed, and portions of the preliminary inner oxide layer **P240** that are covered by the gap fill insulating pattern **244** may remain as the inner oxide layer **240**. The gap fill insulating pattern **244** may be apart from the direct contact DC with the inner oxide layer **240** therebetween. The inner oxide layer **240** and the gap fill insulating pattern **244** may cover both sidewalls of the lower conductive layer **130** of the bit line BL and both sidewalls of the direct contact DC.

(139) Referring to FIG. **12C**, in a manner similar to that described for the method of forming the carbon-containing oxide layer **142** with reference to FIG. **11H**, a carbon-containing oxide layer **242** conformally covering the exposed surfaces of the result illustrated in FIG. **12B** is formed. The carbon-containing oxide layer **242** may contact the intermediate conductive layer **132**, the upper conductive layer **134**, and the insulating capping pattern **136**.

(140) According to an example embodiment, the carbon-containing oxide layer **2422** may substantially have a constant thickness in the vertical direction (Z direction) on the sidewall of the bit line BL and the sidewall of the insulating capping pattern **136**. In other example embodiments, similar to the forming method of the carbon-containing oxide layer **142A** illustrated in FIG. **3** with reference to FIG. **11H**, the carbon-containing oxide layer **242A** illustrated in FIG. **6** may be formed in the operation in FIG. **12C**, instead of the carbon-containing oxide layer **242**.

(141) Referring to FIG. **12D**, in the same manner as described for the method of forming the plurality of intermediate insulating spacers **146** with reference to FIG. **11K**, the plurality of intermediate insulating spacer **246** covering both sidewalls of each of the plurality of bit lines BL may be formed from the result illustrated in FIG. **12C**. The plurality of intermediate insulating spacers **246** may be apart from the bit line BL and the insulating capping pattern **136** with the carbon-containing oxide layer **242** therebetween.

(142) After the plurality of intermediate insulating spacers **246** are formed, by etching portions of the carbon-containing oxide layer **242** that are continuously exposed from the bottom portions of the plurality of line spaces LS, a portion of the inner oxide layer **240**, a portion of the gap fill insulating pattern **244**, and a portion of the buffer layer **122**, a portion of the substrate **110** and a portion of the buried insulating layer **120** may be exposed through the plurality of line spaces LS.

(143) Referring to FIG. **12E**, the outer insulating spacers **148** covering the plurality of intermediate insulating spacers **246** may be formed in the same manner as the method described with reference to FIG. **11L** in the result illustrated in FIG. **12D**.

(144) Referring to FIG. **12F**, in the result illustrated in FIG. **12E**, in the same manner as the method described with reference to FIG. **11M**, the plurality of insulating fences **149** may be formed in the line space LS between each of the plurality of bit lines BL, the LS may be divided into the plurality of contact spaces CS, and then the plurality of recess spaces R1 communicating with the plurality of contact spaces CS may be formed.

(145) A portion of the active region ACT of the substrate **110**, a portion of the inner oxide layer **240**, and a portion of the gap fill insulating pattern **144** may be exposed through the plurality of recess spaces R1. Because the carbon-containing oxide layer **242** is covered with the outer insulating spacer **148**, the carbon-containing oxide layer **242** may not be exposed in the plurality of contact spaces CS and the plurality of recess spaces R1.

(146) Referring to FIG. **12G**, in the same manner as the method described with reference to FIG. **11N** in the result illustrated in FIG. **12F**, the plurality of contact plugs **150** may be formed between each of the plurality of bit lines BL.

(147) Referring to FIG. **12H**, the plurality of metal silicide layers **172** and the plurality of

conductive landing pads LP may be formed on the resultant product illustrated in FIG. 12G, and after the upper recess space R2 is formed around the plurality of conductive landing pads LP, the insulating layer **180** filling the upper recess space R2 may be formed and then the integrated circuit device **200** illustrated in FIGS. 5A-5C may be manufactured.

(148) According to an example embodiment, after the upper recess space R2 around the plurality of conductive landing pads LP is formed in the process described with reference to FIG. 12H, and before the upper recess space R2 is filled with the insulating layer **180**, at least a portion of the plurality of intermediate insulating spacers **146** may be removed through the upper recess space R2.

(149) In an example, the silicon oxide layer constituting the plurality of intermediate insulating spacers **246** may be completely removed through the upper recess space R2 so that the intermediate insulating spacer **246** is formed as an air spacer.

(150) In another example, to manufacture the integrated circuit device **200B** illustrated in FIG. 7, after the upper recess space R2 is formed in the process described with reference to FIG. 12H, and before the upper recess space R2 is filled with the insulating layer **180**, the air spacer AS2 may be formed by removing a portion of the silicon oxide layer constituting the plurality of intermediate insulating spacer **246** through the upper recess space R2, and the intermediate insulating spacer pattern P1 including a remaining portion of the intermediate insulating spacer **246** at the bottom portion of the air spacer AS2 may be maintained.

(151) FIGS. 13A and 13B are cross-sectional views illustrating a method of manufacturing an integrated circuit device according to another example embodiment of the inventive concept. In FIGS. 13A and 13B, (a) is a cross-sectional view of some components according to a process sequence of a portion corresponding to a cross-section taken along line A-A' of FIG. 1, (b) is a cross-sectional view of some components according to a process sequence of a portion corresponding to a cross-section taken along line B-B' of FIG. 1, and (c) is an enlarged cross-sectional view of a portion corresponding to a dash-lined region AX in (a). An example manufacturing method of the integrated circuit device **300** illustrated in FIG. 8 are described with reference to FIGS. 13A and 13B. In FIGS. 13A and 13B, the same reference numerals as those in FIGS. 1 through 12H may denote the same members, and descriptions thereof are omitted here.

(152) Referring to FIG. 13A, in a manner similar to the method of forming the inner oxide layer **240** and the gap fill insulating pattern **244** with reference to FIGS. 12A and 12B, the inner oxide layer **340** and the gap fill insulating pattern **344** may be formed. However, in FIGS. 13A and 13B, the inner oxide layer **340** and the gap fill insulating pattern **344** may be formed to cover both sidewalls of the direct contact DC at a level lower than the top surface of the buffer layer **122**. The inner oxide layer **340** and the gap fill insulating pattern **344** may not cover both sidewalls of the direct contact DC at a level higher than the top surface of the buffer layer **122**. Both sidewalls of the lower conductive layer **130** of each of the plurality of bit lines BL may not be covered with the inner oxide layer **340** and the gap fill insulating pattern **344**. The inner oxide layer **340** may be between the direct contact DC and the gap fill insulating pattern **344**. The gap fill insulating pattern **344** may be apart from the direct contact DC with the inner oxide layer **340** therebetween.

(153) Referring to FIG. 13B, in the result illustrated in FIG. 13A, in the similar method to the forming method, described with reference to FIGS. 12C and 12D, of the carbon-containing oxide layer **242** and the intermediate insulating spacer **246**, the carbon-containing oxide layer **342** and the intermediate insulating spacer **346** may be formed.

(154) The carbon-containing oxide layer **342** may conformally cover the sidewall of the bit line BL, the sidewall of the insulating capping pattern **136**, the sidewall of the direct contact DC, the top surface of the inner oxide layer **340**, and the top surface of the gap fill insulating pattern **344**. The carbon-containing oxide layer **342** may contact the lower conductive layer **130**, the intermediate conductive layer **132**, and the upper conductive layer **134** of the bit line BL, and the plurality of insulating capping patterns **136**. A more detailed configuration of the carbon-containing

oxide layer **342** may be substantially the same as that described for the carbon-containing oxide layer **142** with reference to FIGS. **2** and **11H**.

(155) According to an example embodiment, the carbon-containing oxide layer **342** may substantially have a constant thickness in the vertical direction (Z direction) on the sidewall of the bit line BL and the sidewall of the insulating capping pattern **136**. In other example embodiments, similar to the forming method of the carbon-containing oxide layer **142A** illustrated in FIG. **3** with reference to FIG. **11H**, the carbon-containing oxide layer **342A** illustrated in FIG. **9** may be formed in the operation in FIG. **13B**, instead of the carbon-containing oxide layer **342**. Next, by performing processes similar to those described with reference to FIGS. **12E** through **12H**, the integrated circuit device **300** illustrated in FIG. **8** may be manufactured.

(156) To manufacture the integrated circuit device **300B** illustrated in FIG. **10**, in the process described with reference to FIG. **12H**, the air spacer AS3 may be formed by removing a portion of the silicon oxide layer constituting the plurality of intermediate insulating spacers **146B** through the upper recess space R2, and the intermediate insulating spacer pattern P3 including the remaining portion of the silicon oxide layer may remain under the air spacer AS3.

(157) While the inventive concept has been particularly shown and described with reference to example embodiments thereof, it will be understood that various changes in form and details may be made therein without departing from the spirit and scope of the following claims.

Claims

1. An integrated circuit device, comprising: a substrate comprising a plurality of active regions; a device isolation layer on the substrate, the device isolation layer defining the plurality of active regions; a plurality of bit lines spaced apart from each other on the substrate in a first direction, the plurality of bit lines extending in a second direction crossing the first direction, each of the plurality of bit lines comprising a metal layer, an uppermost surface of the metal layer of each of the plurality of bit lines being on a first level of a first line extending in the first line on the plurality of active regions and on the device isolation layer; a direct contact provided between a first active region among the plurality of active regions and a first bit line among the plurality of bit lines; an inner oxide layer contacting a sidewall of the direct contact; and a carbon-containing oxide layer contacting a sidewall of the first bit line, the carbon-containing oxide layer being separate from the inner oxide layer.
2. The integrated circuit device of claim 1, wherein a lowermost surface of the metal layer of each of the plurality of bit lines is on a second level of a second line extending in the first line on the plurality of active regions and on the device isolation layer.
3. The integrated circuit device of claim 1, further comprising: a plurality of contact plugs arranged in a line along the first direction, each of the plurality of contact plugs being provided one by one between two adjacent bit lines among the plurality of bit lines; and a plurality of conductive landing pads on the plurality of contact plugs and connected to the plurality of contact plugs, wherein an uppermost surface of each of the plurality of conductive landing pads is on a third level of a third line extending in the first line.
4. The integrated circuit device of claim 1, further comprising a plurality of insulating capping patterns covering the plurality of bit lines, wherein a first height of each of the plurality of insulating capping patterns is greater than a second height of the metal layer of each of the plurality of bit lines.
5. The integrated circuit device of claim 1, further comprising a plurality of insulating capping patterns covering the plurality of bit lines, wherein a bottom of each of the plurality of insulating capping patterns contacts the uppermost surface of the metal layer.
6. The integrated circuit device of claim 1, further comprising a buffer layer provided between the substrate and the metal layer, the buffer layer vertically overlapping the metal layer, wherein the

buffer layer includes a silicon nitride layer.

7. The integrated circuit device of claim 1, further comprising: a contact plug connected to a second active region adjacent to the first active region among the plurality of active regions, and a gap fill insulating pattern provided between a lower portion of the contact plug and the direct contact, wherein the inner oxide layer is spaced apart from the contact plug with the gap fill insulating pattern therebetween.

8. The integrated circuit device of claim 1, further comprising: a contact plug connected to a second active region adjacent to the first active region among the plurality of active regions, and a gap fill insulating pattern provided between a lower portion of the contact plug and the direct contact, wherein the carbon-containing oxide layer comprises a portion provided between the direct contact and the gap fill insulating pattern.

9. The integrated circuit device of claim 1, wherein the carbon-containing oxide layer includes an SiOC layer.

10. The integrated circuit device of claim 1, further comprising: a contact plug connected to a second active region adjacent to the first active region among the plurality of active regions, a gap fill insulating pattern provided between a lower portion of the contact plug and the direct contact, and an outer insulating spacer covering the sidewall of the first bit line on the gap fill insulating pattern, wherein at least one of the gap fill insulating pattern and the outer insulating spacer includes a silicon nitride layer.

11. The integrated circuit device of claim 1, wherein the metal layer of each of the plurality of bit lines includes tungsten.

12. An integrated circuit device, comprising: a substrate comprising a plurality of active regions; a first bit line and a second bit line spaced apart from each other on the substrate in a first direction, each of the first bit line and the second bit line extending in a second direction crossing the first direction, each of the first bit line and the second bit line comprising a metal layer; a direct contact provided between a first active region among the plurality of active regions and the first bit line; an inner oxide layer contacting a sidewall of the direct contact; and a carbon-containing oxide layer contacting a sidewall of the first bit line, the carbon-containing oxide layer being separate from the inner oxide layer, wherein an uppermost surface of the metal layer of each of the first bit line and the second bit line is on a first level of a first line, and wherein a lowermost surface of the metal layer of each of the first bit line and the second bit line is on a second level of a second line parallel to the first line.

13. The integrated circuit device of claim 12, further comprising: a plurality of contact plugs arranged in a line along the first direction; and a plurality of conductive landing pads on the plurality of contact plugs and connected to the plurality of contact plugs, wherein a first contact plug among the plurality of contact plugs is provided between the first bit line and the second bit line, and the first contact plug is connected to a second active region adjacent to the first active region among the plurality of active regions, and wherein an uppermost surface of each of the plurality of conductive landing pads is on a third level of a third line extending in the first line.

14. The integrated circuit device of claim 12, further comprising: a first insulating capping pattern provided on the first bit line in a vertical direction; and a second insulating capping pattern provided on the second bit line in the vertical direction, wherein a first height of each of the first and second insulating capping patterns is greater than a second height of the metal layer of each of the first and second bit lines in the vertical direction.

15. The integrated circuit device of claim 12, further comprising: a first buffer layer provided between the substrate and the metal layer of the first bit line in a vertical direction; and a second buffer layer provided between the substrate and the metal layer of the second bit line in the vertical direction, wherein each of the first and second buffer layers includes a silicon nitride layer.

16. The integrated circuit device of claim 12, further comprising: a contact plug connected to a second active region adjacent to the first active region among the plurality of active regions, the

contact plug being provided between the first bit line and the second bit line; a gap fill insulating pattern provided between a lower portion of the contact plug and the direct contact; a first outer insulating spacer provided between the first bit line and the contact plug; and a second outer insulating spacer provided between the second bit line and the contact plug, wherein each of the gap fill insulating pattern, the first outer insulating spacer, and the second outer insulating spacer includes a silicon nitride layer.

17. An integrated circuit device, comprising: a substrate comprising a plurality of active regions; a device isolation layer on the substrate, the device isolation layer defining the plurality of active regions; a plurality of bit lines spaced apart from each other on the substrate in a first direction, the plurality of bit lines extending in a second direction crossing the first direction, each of the plurality of bit lines comprising a metal layer; a direct contact provided between a first active region among the plurality of active regions and a first bit line among the plurality of bit lines; a plurality of contact plugs arranged in a line along the first direction, each of the plurality of contact plugs being provided one by one between two adjacent bit lines among the plurality of bit lines and extending on the substrate in a vertical direction perpendicular to the first and second directions, a first contact plug among the plurality of contact plugs being connected to a second active region adjacent to the first active region among the plurality of active regions; and a spacer structure provided between the first bit line and the first contact plug, the spacer structure comprising an inner oxide layer contacting a sidewall of the direct contact, and a carbon-containing oxide layer contacting a sidewall of the metal layer of the first bit line, the carbon-containing oxide layer being separate from the inner oxide layer, wherein an uppermost surface of the metal layer of each of the plurality of bit lines is on a first level of a first line, and wherein a lowermost surface of the metal layer of each of the plurality of bit lines is on a second level of a second line parallel to the first line.

18. The integrated circuit device of claim 17, further comprising a plurality of conductive landing pads on the plurality of contact plugs and connected to the plurality of contact plugs, wherein an uppermost surface of each of the plurality of conductive landing pads is on a third level of a third line extending in the first line.

19. The integrated circuit device of claim 17, further comprising: a plurality of insulating capping patterns each vertically overlapping the metal layer of each of the plurality of bit lines; and a plurality of buffer layers each provided between the substrate and the metal layer of each the plurality of bit lines, each of the plurality of buffer layers vertically overlapping the metal layer of each of the plurality of bit lines, wherein a first height of each of the plurality of insulating capping patterns is greater than a second height of the metal layer of each of the plurality of bit lines.

20. The integrated circuit device of claim 17, wherein the metal layer of each of the plurality of bit lines includes tungsten, wherein the carbon-containing oxide layer includes an SiOC layer, and wherein the spacer structure further comprises an outer insulating spacer covering the sidewall of the metal layer, the outer insulating spacer including a silicon nitride layer.
